Course: ENCN423/ENCN627
Module 10: Fuel Practice Questions

Question 1

Which of the following statements regarding green hydrogen production from water electrolysis is correct?

(a) Higher electrolyser efficiency reduces electricity consumption per unit of hydrogen produced

(b) Electrolyser efficiency has no effect on hydrogen production costs

(c) Hydrogen storage is usually the dominant contributor to levelised costs of hydrogen

(d) Electrolysis converts hydrogen directly into electricity

Question 2

In most green hydrogen production systems, which component contributes the largest share of hydrogen costs?

(a) Water cost

(b) Electricity cost

(c) Electrolysis CAPEX

(d) Land lease cost

Question 3

How e-fuel production help reduce renewable energy curtailment in high-renewable power systems?

(a) E-fuel plants can convert excess renewable electricity into storable chemical energy

(b) E-fuels eliminate the need for electrolysis

(c) Renewable generators stop producing electricity during low demand periods

(d) E-fuels consume fossil fuels during production

Question 4

A renewable-energy-based hydrogen production system supplies green hydrogen for the production of three electrofuels: e-ammonia, e-methanol, and e-Fischer–Tropsch liquids (e-FTL). Using the techno-economic data provided below, calculate the levelised cost of hydrogen and subsequently determine the levelised costs of the three electrofuels.

Table 1: E-fuel plant techno-economic parameters

Fuel	hydrogen	e-methanol	e-ammonia	e-FTL
Plant capacity	10 MW_{el}	5 MW_{fuel}	4 MW_{fuel}	3 MW_{fuel}
Plant CAPEX ($/kW)	1000	1200	900	1800
Fixed O&M cost (% annual CAPEX)	4%	4%	4%	4%
Plant efficiency (%)	70	–	–	–
Plant lifetime (Years)	30	30	30	30
Discount rate (%)	7	7	7	7
Annual operating hours	8000	6000	6000	6000
Power req. (MWh/MWh_{fuel})	—	0.03	0.14	0.05
CO_2 req. (tCO_2/MWh_{fuel})	—	0.25	—	0.26
H2 consumption (MWh/MWh_{fuel})	—	1.2	1.1	1.5
Electricity price ($/MWh)	40	40	40	40
CO_2 price ($\$/tCO_2$)	—	120	—	120